Zihua Liu

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Professional Summary

Machine Learning Engineer with a Ph.D. in computer vision and 5+ years of combined research and engineering experience since 2021. Build production perception systems for logistics digital twins and GPU workflows on AWS, delivering reusable ML capabilities to customer teams in Japan and overseas.

Research & Engineering Experience

Woven by Toyota, Inc. Tokyo, Japan
• Machine Learning Engineer — Digital Genba Aug 2025 – Present

- Built the multimodal perception pipeline for Digital Genba's logistics digital twin, from data capture and 3D Gaussian Splatting (3DGS) reconstruction to multi-camera detection, tracking, and VLM-based behavior analysis and reasoning.
- Built production MLOps workflows with Metaflow, AWS Batch, Step Functions, and Docker. Implemented resumable GPU jobs, reproducible run configurations, and artifact publishing to S3 and Lance.
- Delivered reusable vision and multimodal solutions to multiple customer teams in Japan and overseas, including Toyota teams, by translating logistics and industrial-site requirements into shared platform components.
- Designed the model delivery workflow from annotation and dataset versioning through training, evaluation, inference, visualization, and cloud deployment.

Sony Semiconductor Solutions Tokyo, Japan
• R&D Research Intern Jan 2025 – Mar 2025

- Trained a multi-view diffusion model on the GAIA GPU cluster to generate synthetic street-view and bird's-eye-view data for autonomous perception. Built a synchronized pipeline for LiDAR, images, 3D boxes, and text.

Preferred Networks Tokyo, Japan
• R&D Research Intern Aug 2024 – Oct 2024

- Developed a controllable diffusion model for background generation and validated its cross-dataset generalization by benchmarking against established methods across datasets and diverse foreground styles.

SenseTime Japan Tokyo, Japan
• Computer Vision Research Intern Jun 2023 – Nov 2023

- Co-developed VSRD (CVPR 2024), raising KITTI-360 Hard BEV AP@0.3 from 37.34 to 41.28 versus Autolabels. Generated 3D pseudo-labels from 2D masks, eliminating manual 3D box annotation for training.

Megvii Research Shanghai, China
• Computer Vision Research Intern Feb 2022 – Apr 2022

- Increased parking-slot detection recall by 10% through geometric constraints and targeted augmentation for occlusion and image-edge cases. Megvii adopted the model in its automatic-parking perception pipeline.

Technical Skills

- **MLOps & Cloud:** Metaflow, AWS Batch, Step Functions, S3, Lance, Docker, Linux, Git, CI/CD, dataset versioning, model evaluation, GPU inference and deployment
- **Programming & ML:** Python, C++, CUDA, PyTorch, distributed training, vLLM, VLMs, diffusion models, object detection, segmentation, ReID, multi-camera tracking, NeRF, 3D Gaussian Splatting
- **Research & Evaluation:** Multi-view geometry, stereo depth, weak supervision, 3D reconstruction, diffusion modeling, benchmark evaluation, and ablation studies
- **Languages:** English, Japanese, Chinese

Education

Institute of Science Tokyo (Tokyo Institute of Technology) Tokyo, Japan
• Ph.D. in Engineering, Computer Vision and Deep Learning Mar 2023 – Mar 2026
• M.Eng. in Engineering Apr 2021 – Mar 2023

South China University of Technology Guangzhou, China
• B.Eng. in Engineering Sep 2016 – Jun 2020

Selected Publications

- **Zihua Liu** and Masatoshi Okutomi, "StereoSplat+: Feed-Forward Stereo Gaussian Splatting with Diffusion-Assisted Progressive Inference," *IROS*, 2026.
- **Zihua Liu** et al., "DMS: Diffusion-Based Multi-Baseline Stereo Generation for Improving Self-Supervised Depth Estimation," *ICCV Workshops (AIM, Oral),* 2025.
- **Zihua Liu** et al., "VSRD: Volumetric Silhouette Rendering for Weakly Supervised 3D Object Detection,"*CVPR*, 2024.
- **Zihua Liu** et al., "CFDNet: A Generalizable Foggy Stereo Matching Network with Contrastive Feature Distillation," *ICRA*, 2024.

Honors & Awards

- Best Paper Award, Symposium on Sensing via Image Information (SSII), 2026; Best Student Paper Award, International Conference on Multimedia Modeling (MMM), 2025
- Tokyo Tech Advanced Human Resource Development Fellowship, 2023–2026; SCIE Student Award, 2023